The Alaoglu–Erdős Integer-Exponent Conjecture

Dyadic Cyclotomic Defects, Fresh Prime Supports, and a Critical Matrix-Logarithm Target

Research continuation prepared by GPT-5.6 Pro

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Mathematical status. This report proves several new exact lemmas and a sharp no-go theorem, but it does *not* prove the Alaoglu–Erdős conjecture. The final support-sensitive determinant estimate identified below remains open.

Abstract

Assume that $x > 0$ and that

$$M = 2^x \in \mathbb{Z}, \qquad A = 3^x \in \mathbb{Z}.$$

Writing $\vartheta = \log_2 3$, so that $A = M^\vartheta$, we introduce moving logarithmic determinants built from $M^n \pm 1$ and $A^n \pm 1$. Their signs are exact, their magnitudes are exponentially small, and they satisfy positive telescoping partitions indexed by every prime-power cyclotomic ladder. In the dyadic ladder the factors $M^{2^j} + 1$ have pairwise disjoint odd prime supports; every odd divisor has order exactly 2^{j+1}, giving an infinite family of fresh-support, almost rank-one 2×2 matrices of logarithms. The defect has the exact interpretation of a Rényi entropy and converges at the critical normalized slope $\log_3 2$.

We then prove a sharp contact ceiling: no fixed monic polynomial transfer can create a smaller determinant, relative to logarithmic height, than the dyadic binomial family; within the cyclotomic family, powers of two are uniquely optimal. Finally, in the rank-three external-support branch inherited from the unified report, we derive a new integer certificate

$$D = a_2 r^2 - (m_2 - a_3) rs - m_3 s^2 \neq 0$$

and an exact factorization connecting its sign to the rational approximation $r \log 3 - s \log 2$.

These results isolate a precise possible closing theorem: a quantitative lower bound for matrices of logarithms that improves at the critical slope when the primitive output pair has external prime support. Qualitative Baker and Six Exponentials theorems do not supply that improvement, and ordinary integer solutions exhibit the same analytic and cyclotomic identities. The report therefore separates rigorous progress from the still missing arithmetic input and gives a concrete Lean formalization plan.

Contents

1 Executive assessment

Executive assessment

The new route is not another reformulation of the original determinant. It manufactures an infinite sequence of *nonzero*, sign-controlled determinants whose columns acquire fresh prime support at every dyadic level while their distance from rank one decays doubly exponentially in the level. The route reaches a sharp boundary: the same critical decay occurs for every known integer solution $x \in \mathbb{Z}_{>0}$. Consequently, the final theorem must distinguish structural pairs $(2^m, 3^m)$ from simultaneously power-primitive pairs with primes outside $\{2, 3\}$.

The principal rigorous contributions of this continuation are as follows.

1. **Signed shifted determinants.** The two families

$$(\log 3)\log(M^n \pm 1) - (\log 2)\log(A^n \pm 1)$$

   have opposite fixed signs, explicit two-sided bounds, and exact asymptotics.

2. **Positive prime-power partitions.** For every prime p, the minus-defect splits into a convergent sum of positive logarithmic determinants of $\Phi_{p^j}(M^n)$ and $\Phi_{p^j}(A^n)$. The partition exponentiates to an exact infinite product independent of p.

3. **Entropy identity.** Every geometric-quotient defect is exactly $(\vartheta - 1)\log 2$ times a Rényi entropy of a finite geometric distribution.

4. **Dyadic support theorem.** The odd parts of $T^{2^j} + 1$ are pairwise coprime. Every odd prime divisor is private to its level and is congruent to 1 modulo 2^{j+1}. Hence the associated logarithms are rationally independent.

5. **Fresh-support near-rank-one matrices.** The matrix with columns $(\log 2, \log 3)^T$ and $(\log(M^{2^j} + 1), \log(A^{2^j} + 1))^T$ has a nonzero determinant with exact sign, fresh arithmetic support, and critical decay rate $\log_3 2$ relative to height.

6. **Polynomial contact ceiling.** A fixed monic polynomial of degree d whose first lower term is cT^{d-r} has normalized contact slope $r/(d\vartheta)$. This never exceeds $1/\vartheta$; equality forces a binomial $T^d + c$. For cyclotomic polynomials the slope is $1/(\vartheta\varphi(\operatorname{rad} m))$, so powers of two are uniquely optimal.

7. **Rank-three integer certificate.** When all external prime exponents lie on one ray, the quadratic rank condition gives an explicit nonzero integer D, exact sign information, and a lower bound for $|r\vartheta - s|$.

Do not overclaim

None of the preceding statements rules out a noninteger x. In particular, the private prime theorem does not control overlap between the M-ladder and the A-ladder, and Six Exponentials gives transcendence without a quantitative lower bound of the required critical strength.

2 Source audit, current claims, and nonduplication

2.1 The attached unified report

The source basis was the attached `alaoglu-erdos-unified-report.tex.gz`, a gzip-compressed LaTeX file. After decompression it contains 20,483 lines (974,791 bytes).

The unified report already develops, among much else, the four-exponentials reduction, exact conditional semigroup structure, primitive-generator reductions, valuation-lattice geometry, determinant and interpolation hierarchies, p-adic and matrix-coefficient approaches, entropy and automata viewpoints, and a quadratic-rank classification. This continuation therefore does *not* repeat those surveys or their proofs. It imports only the minimal setup in Section 3 and the rank-three external-ray conclusion in Section 10.

A targeted text audit found no treatment in the source of the exact shifted families $M^n \pm 1$, their positive cyclotomic defect partitions, the dyadic pairwise-gcd/private- prime theorem used here, or the fixed-polynomial contact ceiling. The present report is organized around those new items.

2.2 Audit of the recent public claims

The user's motivating claims were checked against public sources available on 22 August 2026. The careful versions are recorded in Table 1.

Claim	Publicly supported statement	Necessary qualification
Jacobian conjecture	A public Lean repository verifies an explicit polynomial map in three variables with constant Jacobian determinant -2 and a three-point fiber. This refutes the usual conjecture in dimension 3, hence in every dimension at least 3.	The two-dimensional conjecture remains open. "Resolved" without this dimensional qualification is misleading.
Hadamard orders below 2000	OEIS A007299 records 2026 constructions for all twelve previously unknown multiples of four at most 2000: 668, 716, 892, 1132, 1244, 1388, 1436, 1676, 1772, 1916, 1948, and 1964.	The general Hadamard conjecture remains open.
Elliptic curve of rank at least 30	The ICARM leaderboard lists curve #273, submitted 20 August 2026, with a certified lower bound ≥ 30 and thirty independent rational points.	The unconditional statement is a lower bound. The page notes exact rank 30 only under GRH and BSD.
Alaoglu–Erdős breakthrough	Targeted searches of current news, arXiv, and public code repositories did not locate a public preprint, proof, or Lean repository matching the private claim.	Absence from public search is not evidence that private work does not exist. There was nothing public to inspect or reproduce as of the access date.

Table 1: Current-claim audit. Source URLs and access dates appear in the bibliography.

The Jacobian repository describes eleven Lean files with no `sorry` and verifies the constant determinant and explicit collision. ScienceDaily's republication of a The Conversation article attributes the discovery to an AI-assisted workflow involving Claude Fable 5, while explicitly saying that dimension two remains open. OEIS attributes the twelve Hadamard constructions to Alpöge, Philippe Voinov, Saul Reynolds-Haertle, and Claude. The ICARM commentary attributes the rank-30 construction to Claude with Levent Alpöge and Ava Howell. These attributions are reported as source statements rather than independently reconstructed histories.

2.3 Trust ledger

Status and trust boundary

- Statements proved from elementary real analysis, unique factorization, and finite algebra are marked NEW and supplied with proofs.
- Baker's homogeneous linear-independence theorem and Six Exponentials are marked CLASSICAL and cited rather than reproved.
- The rank-three external-ray reduction is marked INHERITED because it comes from the attached unified report.
- The quantitative support-sensitive matrix-logarithm estimate is marked OPEN. No numerical experiment or heuristic is promoted to a proof.

3 Minimal inherited setup

The case $x < 0$ is impossible because $0 < 2^x < 1$ cannot be an integer. The case $x = 0$ is already integral. Hence throughout the main argument we assume

$$x > 0, \qquad M = 2^x \in \mathbb{Z}_{>1}, \qquad A = 3^x \in \mathbb{Z}_{>1}. \tag{1}$$

Put

$$X = \log 2, \qquad Y = \log 3, \qquad \vartheta = \frac{Y}{X} = \log_2 3, \qquad \alpha = \frac{1}{\vartheta} = \frac{X}{Y}. \tag{2}$$

Then

$$\log A = \vartheta \log M, \qquad A = M^{\vartheta}. \tag{3}$$

Unique factorization shows that $\vartheta \notin \mathbb{Q}$. In fact ϑ is transcendental: if it were algebraic irrational, the Gelfond–Schneider theorem applied to $2^{\vartheta} = 3$ would give a contradiction.

For positive real numbers B, C, define the oriented logarithmic determinant

$$\mathscr{D}(B, C) = Y \log B - X \log C = X \log\left(\frac{B^{\vartheta}}{C}\right). \tag{4}$$

It is additive under componentwise multiplication:

$$\mathscr{D}(B_1B_2, C_1C_2) = \mathscr{D}(B_1, C_1) + \mathscr{D}(B_2, C_2). \tag{5}$$

The fundamental relation is simply

$$\mathscr{D}(M, A) = 0, \qquad \mathscr{D}(M^n, A^n) = 0 \quad (n \geq 1). \tag{6}$$

The attached unified report proves or records the following facts that will only be used for context later.

- A hypothetical noninteger solution is transcendental and gives a primitive output pair after passing to the least positive noninteger generator.
- Its prime-exponent logarithmic tensor has quadratic rank 3 or 4; rank at most 2 is excluded by Baker's theorem.
- In rank 3, all exponent vectors at primes outside $\{2, 3\}$ lie on one rational ray. This is the sole inherited structural input used in Section 10.

A pair (M, A) will be called *simultaneously power-primitive* when there is no integer $d \geq 2$ for which both M and A are perfect dth powers. Equivalently, the greatest common divisor of all nonzero prime valuations occurring in M and A is 1. The least nonintegral generator's output pair has this property. We say that the pair has *external support* when a prime outside $\{2, 3\}$ divides MA.

No statement in the next six sections requires the least-generator hypothesis: the exact defect identities hold for every pair satisfying (1).

4 Signed shifted logarithmic defects

The exact relation (6) is unchanged by taking powers. The first new step is to perturb those powers by ± 1 before applying the logarithmic determinant.

Definition 4.1. For $n \geq 1$, define

$$\Delta_n^+ = \mathscr{D}(M^n + 1, A^n + 1), \tag{7}$$

$$\Delta_n^- = \mathscr{D}(M^n - 1, A^n - 1), \tag{8}$$

$$E_n = -\Delta_n^-. \tag{9}$$

Theorem 4.2 (Sign, monotonicity, asymptotics, and dyadic conservation)**.** *For every $n \geq 1$ one has*

$$\Delta_n^- < 0 < \Delta_n^+, \qquad E_n > 0. \tag{10}$$

The sequence $(E_n)_{n\geq 1}$ is strictly decreasing and tends to 0. Moreover,

$$E_n = YM^{-n} - XA^{-n} + O(M^{-2n}), \tag{11}$$

$$\Delta_n^+ = YM^{-n} - XA^{-n} + O(M^{-2n}), \tag{12}$$

so that

$$\lim_{n\to\infty} E_n^{1/n} = \lim_{n\to\infty} (\Delta_n^+)^{1/n} = M^{-1}. \tag{13}$$

Finally,

$$\Delta_n^+ + \Delta_n^- = \Delta_{2n}^-, \qquad \Delta_n^+ = E_n - E_{2n}, \tag{14}$$

and hence the positive dyadic partition

$$E_n = \sum_{j=0}^{\infty} \Delta_{2^j n}^+. \tag{15}$$

Proof. Set $u = M^{-n} \in (0, 1)$. By (3), $A^{-n} = u^\vartheta$. Cancellation of the leading power terms gives the exact formulas

$$\frac{\Delta_n^+}{X} = \vartheta \log(1 + u) - \log(1 + u^\vartheta), \tag{16}$$

$$\frac{\Delta_n^-}{X} = \vartheta \log(1-u) - \log(1-u^\vartheta). \tag{17}$$

Since $\vartheta > 1$ and $u > 0$,

$$(1+u)^\vartheta > 1 + u^\vartheta,$$

which proves $\Delta_n^+ > 0$. Also

$$u^\vartheta + (1-u)^\vartheta < u + (1-u) = 1,$$

so $(1-u)^\vartheta < 1 - u^\vartheta$ and $\Delta_n^- < 0$.

For monotonicity, write

$$\frac{E_n}{X} = e(u) := -\vartheta \log(1-u) + \log(1-u^\vartheta).$$

A direct derivative calculation yields

$$e'(u) = \frac{\vartheta(1-u^{\vartheta-1})}{(1-u)(1-u^\vartheta)} > 0. \tag{18}$$

As n increases, $u = M^{-n}$ strictly decreases, and therefore E_n strictly decreases. The limit is 0 by (17).

Taylor expansion at $u = 0$ gives

$$e(u) = \vartheta u - u^\vartheta + O(u^2), \qquad \frac{\Delta_n^+}{X} = \vartheta u - u^\vartheta + O(u^2),$$

because $\vartheta > 1$ and $2\vartheta > 2$. Multiplying by X, and using $X\vartheta = Y$, proves (11)–(12); the root limits follow.

Finally, add the defining logarithms and use $(M^n - 1)(M^n + 1) = M^{2n} - 1$, and similarly for A:

$$\Delta_n^- + \Delta_n^+ = \Delta_{2n}^-.$$

This is (14). Iteration gives

$$E_n = \sum_{j=0}^{J} \Delta_{2^j n}^+ + E_{2^{J+1} n},$$

and the remainder tends to 0. □

The preceding asymptotics can be replaced by explicit inequalities.

Proposition 4.3 (Sharp elementary two-sided bound). *For every $n \geq 1$,*

$$Y M^{-n} \frac{1-(M/A)^n}{(1+M^{-n})(1+A^{-n})} \leq \Delta_n^+ < Y M^{-n}. \tag{19}$$

Consequently,

$$\frac{\Delta_n^+ M^n}{Y} \longrightarrow 1. \tag{20}$$

Proof. Let

$$f(s) = \vartheta \log(1+s) - \log(1+s^{\vartheta}).$$

Then

$$f'(s) = \frac{\vartheta(1-s^{\vartheta-1})}{(1+s)(1+s^{\vartheta})}. \tag{21}$$

The numerator decreases and the denominator increases on $(0,1)$, so f' decreases. Thus, for $0 < u < 1$,

$$uf'(u) \leq \int_0^u f'(s)\,ds = f(u) < \vartheta u.$$

Substitute $u = M^{-n}$, note that $u^{\vartheta} = A^{-n}$ and $u^{\vartheta-1} = (M/A)^n$, and multiply by X. □

Remark 4.4 (A precise analytic picture)**.** The plus and minus defects have the same first-order magnitude, but the dyadic identity shows that the plus defect is the amount by which the positive "minus energy" drops when the scale doubles. Thus E_n is not merely asymptotic to a geometric sequence; it is exactly the total mass of a positive dyadic cascade.

5 Geometric quotients, Rényi entropy, and cyclotomic partitions

The dyadic identity is one instance of a prime-power factorization. The correct finite object is the geometric quotient

$$Q_{k,n}(T) = \frac{T^{kn}-1}{T^n-1} = \sum_{\ell=0}^{k-1} T^{\ell n} \qquad (k \geq 2, n \geq 1). \tag{22}$$

Definition 5.1. Define the geometric-quotient defect

$$\mathcal{G}_{k,n} = \mathscr{D}(Q_{k,n}(M), Q_{k,n}(A)). \tag{23}$$

Theorem 5.2 (Exact positive quotient law)**.** *For all $k \geq 2$ and $n \geq 1$,*

$$\mathcal{G}_{k,n} = \Delta^-_{kn} - \Delta^-_n = E_n - E_{kn} > 0. \tag{24}$$

For fixed n, the sequence $\mathcal{G}_{k,n}$ increases strictly with k and

$$\lim_{k\to\infty} \mathcal{G}_{k,n} = E_n. \tag{25}$$

Proof. Apply the additivity (5) to the quotient in (22). This gives the first equality in (24); the remainder follows from the strict decrease of E_n. □

There is also a direct convexity proof with a useful information-theoretic interpretation.

Theorem 5.3 (Exact Rényi-entropy identity)**.** *Let*

$$p_\ell = \frac{M^{\ell n}}{Q_{k,n}(M)} \qquad (0 \leq \ell < k). \tag{26}$$

Then $(p_0, \ldots, p_{k-1})$ is a probability vector and

$$\frac{\mathcal{G}_{k,n}}{X} = -\log \sum_{\ell=0}^{k-1} p_\ell^{\vartheta} = (\vartheta - 1) H_{\vartheta}(p_0, \ldots, p_{k-1}), \tag{27}$$

where

$$H_\vartheta(p) = \frac{1}{1-\vartheta} \log \sum_\ell p_\ell^\vartheta$$

is the Rényi entropy of order ϑ. In particular positivity is strict because the probability vector has at least two positive coordinates.

Proof. Let $S = \sum_{\ell=0}^{k-1} M^{\ell n} = Q_{k,n}(M)$. Since $A^{\ell n} = (M^{\ell n})^\vartheta$,

$$\begin{aligned} \frac{\mathcal{G}_{k,n}}{X} &= \vartheta \log S - \log \sum_{\ell=0}^{k-1} (M^{\ell n})^\vartheta \\ &= -\log \sum_{\ell=0}^{k-1} \left(\frac{M^{\ell n}}{S}\right)^\vartheta . \end{aligned}$$

This is exactly (27). □

Remark 5.4. The entropy identity does not add randomness to the problem. It packages a deterministic strict ℓ^1–ℓ^ϑ inequality. Its value is that every prime-power cyclotomic increment below becomes a positive entropy packet with an exact total budget.

For a prime p and $j \geq 1$, define

$$\mathcal{C}_{p^j,n} = \mathscr{D}\big(\Phi_{p^j}(M^n), \Phi_{p^j}(A^n)\big). \tag{28}$$

The prime-power identity

$$\Phi_{p^j}(T^n) = \frac{T^{p^j n} - 1}{T^{p^{j-1} n} - 1} = \sum_{\ell=0}^{p-1} T^{\ell p^{j-1} n} \tag{29}$$

places these factors inside Theorem 5.2.

Theorem 5.5 (Positive cyclotomic ladder)**.** *For every prime p, every $j \geq 1$, and every $n \geq 1$,*

$$\mathcal{C}_{p^j,n} = E_{p^{j-1}n} - E_{p^j n} > 0. \tag{30}$$

Moreover,

$$\mathcal{C}_{p^j,n} \sim Y M^{-p^{j-1}n} \qquad (j \to \infty), \tag{31}$$

and one has the exact positive partition

$$E_n = \sum_{j=1}^{\infty} \mathcal{C}_{p^j,n}. \tag{32}$$

Equivalently,

$$\prod_{j=1}^{\infty} \frac{\Phi_{p^j}(M^n)^\vartheta}{\Phi_{p^j}(A^n)} = \frac{1 - A^{-n}}{(1 - M^{-n})^\vartheta}. \tag{33}$$

The sum and product are independent of the chosen prime p.

Proof. Equation (30) follows by applying (5) to (29). Positivity follows from the decrease of E_n, or directly from

$$\left(\sum_{\ell=0}^{p-1} y_\ell\right)^{\vartheta} > \sum_{\ell=0}^{p-1} y_\ell^{\vartheta}$$

for $y_\ell = M^{\ell p^{j-1} n} > 0$. The asymptotic follows from (11), because $E_{p^j n} = o(E_{p^{j-1} n})$. Summation telescopes and the tail tends to zero, proving (32). Divide by X and exponentiate to obtain (33). ☐

Corollary 5.6 (Cross-prime conservation)**.** *For any primes p, q and every $n \geq 1$,*

$$\sum_{j\geq 1} \mathcal{C}_{p^j,n} = \sum_{j\geq 1} \mathcal{C}_{q^j,n} = E_n. \tag{34}$$

New rigorous result in this continuation

The identities (32)–(34) create infinitely many positive decompositions of the same small determinant. Each decomposition has a distinct arithmetic support pattern. The dyadic version is strongest because its factors have pairwise disjoint odd prime supports and, as proved later, it also attains the optimal height-normalized contact slope among all fixed cyclotomic transfers.

6 Dyadic private primes and multiplicative independence

Set

$$W_j(T) = T^{2^j} + 1 \qquad (T > 1,\ j \geq 1). \tag{35}$$

The following elementary fact is stronger than merely invoking a primitive-divisor theorem.

Theorem 6.1 (Exact dyadic gcd law)**.** *For integers $T > 1$ and distinct $i, j \geq 1$,*

$$\gcd(W_i(T), W_j(T)) = \begin{cases} 1, & T \text{ even}, \\ 2, & T \text{ odd}. \end{cases} \tag{36}$$

Thus the odd parts of the $W_j(T)$ are pairwise coprime.

Proof. Assume $i < j$ and let d divide both factors. Modulo d, $T^{2^i} \equiv -1$. Raising to the even power 2^{j-i} gives $T^{2^j} \equiv 1$, while the second divisibility gives $T^{2^j} \equiv -1$. Hence $d \mid 2$. Parity now gives the two cases. ☐

Theorem 6.2 (Private order and congruence)**.** *For every $T > 1$ and $j \geq 1$, the integer $W_j(T)$ has an odd prime divisor. Every odd prime $q \mid W_j(T)$ satisfies*

$$\operatorname{ord}_q(T) = 2^{j+1}, \qquad q \equiv 1 \pmod{2^{j+1}}. \tag{37}$$

In particular, q divides no $W_i(T)$ with $i \neq j$ and $q \geq 2^{j+1} + 1$.

Proof. If T is even, $W_j(T) > 1$ is odd. If T is odd, then $T^{2^j} \equiv 1 \pmod 8$, so $W_j(T) \equiv 2 \pmod 8$ and $W_j(T) > 2$; it therefore has an odd prime divisor q. For such a divisor, $T^{2^j} \equiv -1 \pmod q$. The order divides 2^{j+1} but does not divide 2^j, so it is exactly 2^{j+1}. Fermat's theorem gives the congruence for q. Privacy also follows from Theorem 6.1, or directly from the order. □

Corollary 6.3 (Multiplicative and logarithmic independence)**.** *For every integer $T > 1$, the family $\{W_j(T) : j \geq 1\}$ is multiplicatively independent. Consequently, the real numbers $\{\log W_j(T) : j \geq 1\}$ are linearly independent over $\mathbb{Q}$.*

Proof. In a finite relation $\prod_j W_j(T)^{c_j} = 1$ with $c_j \in \mathbb{Z}$, choose an index with $c_j \neq 0$ and an odd prime divisor private to that index. Its valuation forces $c_j = 0$, a contradiction. Clearing denominators converts a rational linear relation among the logs into such a multiplicative relation. □

The congruence in (37) gives a quantitative, though modest, support budget.

Corollary 6.4 (Support growth)**.** *For each $J \geq 1$, the product $\prod_{j=1}^{J} W_j(T)$ has at least J distinct odd prime divisors q_j with*

$$q_j \geq 2^{j+1} + 1. \tag{38}$$

Hence

$$\sum_{j=1}^{J} \log q_j \geq \sum_{j=1}^{J} \log(2^{j+1} + 1) = \frac{\log 2}{2} J^2 + O(J). \tag{39}$$

Apply these statements to

$$U_j = M^{2^j} + 1, \qquad V_j = A^{2^j} + 1 \qquad (j \geq 1). \tag{40}$$

Because every private odd prime is 1 modulo 4, it is neither 2 nor 3.

Corollary 6.5 (Independence with the base logs)**.** *Each of the two infinite families*

$$\{\log 2, \log 3, \log U_1, \log U_2, \ldots\}, \qquad \{\log 2, \log 3, \log V_1, \log V_2, \ldots\} \tag{41}$$

is linearly independent over $\mathbb{Q}$ in every finite subfamily.

Proof. A private odd prime at each dyadic level first kills the coefficient of the corresponding $\log U_j$ or $\log V_j$. The remaining relation between $\log 2$ and $\log 3$ is trivial by unique factorization. □

Remark 6.6 (What is not proved)**.** The private prime for U_j is private only relative to the other U_i; it may divide a factor V_k. No cross-family disjointness is asserted. Controlling the intersections $\operatorname{supp}(U_j) \cap \operatorname{supp}(V_k)$ is one possible adelic continuation of this route.

7 Fresh-support matrices approaching rank one

Let $r_j = 2^j$. Form the logarithm matrix

$$L_j = \begin{pmatrix} X & \log U_j \\ Y & \log V_j \end{pmatrix} \qquad (j \geq 1), \tag{42}$$

and orient its small determinant by

$$\Gamma_j = Y \log U_j - X \log V_j = -\det L_j. \tag{43}$$

Thus $\Gamma_j = \Delta^+_{r_j}$.

Theorem 7.1 (Fresh-support near-rank-one theorem)**.** *For every $j \geq 1$:*

(i) $\det L_j < 0$, *so* L_j *has rank* 2*;*

(ii) *the second entries* $\log U_j$ *(respectively* $\log V_j$*) acquire a private odd prime support at every level;*

(iii) *the determinant satisfies*

$$Y M^{-r_j} \frac{1 - (M/A)^{r_j}}{(1 + M^{-r_j})(1 + A^{-r_j})} \leq \Gamma_j < Y M^{-r_j}; \tag{44}$$

(iv) *in particular,*

$$\Gamma_j \sim Y M^{-r_j}; \tag{45}$$

(v) *if* $H_j = \log V_j$, *then*

$$\lim_{j \to \infty} \frac{-\log \Gamma_j}{H_j} = \frac{\log M}{\log A} = \frac{X}{Y} = \alpha; \tag{46}$$

(vi) *the relative power approximation is exact:*

$$\frac{U_j^\vartheta}{V_j} = \exp\!\left(\frac{\Gamma_j}{X}\right) > 1, \qquad \frac{U_j^\vartheta}{V_j} - 1 \sim \vartheta M^{-r_j}. \tag{47}$$

Proof. The determinant assertion and the bound are Proposition 4.3 with $n = r_j$. The support statement follows from Theorem 6.2. The asymptotic follows from the bound. Since

$$-\log \Gamma_j = r_j \log M + O(1), \qquad H_j = r_j \log A + o(1),$$

we obtain (46). Finally (4) gives

$$\frac{\Gamma_j}{X} = \log(U_j^\vartheta / V_j),$$

and the final asymptotic follows from $e^t - 1 \sim t$. □

The phrase "near rank one" has an exact Euclidean meaning. Let

$$c = \begin{pmatrix} X \\ Y \end{pmatrix}, \qquad w_j = \begin{pmatrix} \log U_j \\ \log V_j \end{pmatrix}.$$

Proposition 7.2 (Exact distance to the rank-one line)**.** *For every $j \geq 1$,*

$$\operatorname{dist}(w_j, \mathbb{R}c) = \frac{\Gamma_j}{\sqrt{X^2 + Y^2}}. \tag{48}$$

Moreover,

$$\sin \angle(c, w_j) = \frac{\Gamma_j}{\|c\| \, \|w_j\|} \asymp \frac{M^{-r_j}}{r_j}. \tag{49}$$

Proof. The distance from a vector w to the line spanned by a nonzero vector c in $\mathbb{R}^2$ is $|\det(c,w)|/\|c\|$. Here the absolute determinant is Γ_j. The angle formula is the same identity divided by $\|w_j\|$, and $\|w_j\| \asymp r_j$. □

There is a useful finite-product form of the dyadic conservation law. From

$$\prod_{j=0}^{J}(T^{2^j n}+1) = \frac{T^{2^{J+1}n}-1}{T^n-1}$$

and Theorem 5.5, one obtains

$$\prod_{j=0}^{\infty} \frac{(M^{2^j n}+1)^{\vartheta}}{A^{2^j n}+1} = \frac{1-A^{-n}}{(1-M^{-n})^{\vartheta}}. \tag{50}$$

Every factor on the left is greater than 1, and its logarithm is a positive defect. Thus the moving matrices form a genuinely arithmetic, positive approach to rank one, not a sequence with cancellation between signs.

7.1 An exact moving-determinant equivalence

For arbitrary integers $M, A > 1$, not initially assumed to satisfy $Y \log M = X \log A$, define U_j, V_j, L_j as above. Then

$$\Gamma_j = r_j(Y\log M - X\log A) + Y\log(1+M^{-r_j}) - X\log(1+A^{-r_j}). \tag{51}$$

Proposition 7.3 (Decay is equivalent to the original determinant relation)**.** *For fixed integers $M, A > 1$, the following are equivalent:*

(a) $Y \log M - X \log A = 0$*;*

(b) $\Gamma_j \to 0$*;*

(c) $|\Gamma_j| \le C\rho^{r_j}$ *for some* $C > 0$ *and* $0 < \rho < 1$*.*

When these hold, $\Gamma_j > 0$ *for every* j *and the critical slope is* $\log M/\log A$*.*

Proof. Equation (51) shows that if the base determinant is nonzero, then Γ_j/r_j tends to that nonzero value. This excludes (b) and (c). If the base determinant vanishes, the proof of Theorem 7.1 applies verbatim. □

Remark 7.4 (Why this reformulation is useful but not free)**.** The Alaoglu–Erdős conjecture is equivalent to classifying all integer pairs (M, A) for which the second column of these fresh-support matrices approaches the base-log line (or, equivalently, $\det L_j \to 0$): they should be precisely $(M, A) = (2^m, 3^m)$ with $m \in \mathbb{Z}_{>0}$. The equivalence does not itself reduce the logical difficulty. It does, however, replace a single exact zero by an infinite arithmetic sequence carrying new, level-by-level prime supports.

8 Six Exponentials and the critical quantitative target

The dyadic support theorem makes the qualitative Six Exponentials theorem applicable in a particularly clean way.

Theorem 8.1 (Qualitative transcendence along every triple)**.** *Choose any three distinct indices* $i, j, k \geq 1$*. At least one of*

$$U_i^\vartheta, \qquad U_j^\vartheta, \qquad U_k^\vartheta \tag{52}$$

is transcendental.

Proof. The numbers 1 and ϑ are linearly independent over $\mathbb{Q}$, while $\log U_i, \log U_j, \log U_k$ are linearly independent over $\mathbb{Q}$ by Corollary 6.3. Six Exponentials says that at least one of the six numbers

$$\exp(\log U_\ell) = U_\ell, \qquad \exp(\vartheta \log U_\ell) = U_\ell^\vartheta \quad (\ell \in \{i, j, k\})$$

is transcendental. The first three are integers, so at least one of the second three is transcendental. □

This conclusion is compatible with the integer V_j. Indeed $U_j^\vartheta \neq V_j$ by (47); the approximation is only relative.

Proposition 8.2 (Relative closeness but absolute separation)**.** *One has*

$$\frac{U_j^\vartheta - V_j}{V_j} \sim \vartheta M^{-r_j}, \tag{53}$$

$$U_j^\vartheta - V_j \sim \vartheta M^{(\vartheta-1)r_j}. \tag{54}$$

Thus the relative error tends doubly exponentially to zero in j*, while the absolute error tends to infinity.*

Proof. The first statement is (47). Since $V_j \sim A^{r_j} = M^{\vartheta r_j}$, multiplication gives the second. □

Failure audit 8.3. The construction supplies no small absolute distance from U_j^ϑ to an integer: the distinguished integer V_j is already far away in absolute terms, and no claim is made about a different nearest integer. The certified small quantity is the logarithmic determinant, equivalently the relative error. A useful transcendence measure must therefore engage that height-normalized quantity rather than an absolute approximation not furnished by the construction.

Put

$$\lambda_j = \frac{-\log|\det L_j|}{\log V_j}. \tag{55}$$

Under a hypothetical solution, Theorem 7.1 gives $\lambda_j \to \alpha$. This identifies the exact quantitative threshold.

Target conjecture 8.4 (Support-sensitive critical-slope rigidity)**.** *Let* $M, A > 1$ *be integers. Suppose that* (M, A) *is simultaneously power-primitive and has external support. Then there exists* $\varepsilon(M, A) > 0$ *such that, for infinitely many* j*,*

$$|\det L_j| \geq \exp\bigl(-(\alpha - \varepsilon(M, A)) \log V_j\bigr). \tag{56}$$

Equivalently,

$$\liminf_{j\to\infty} \lambda_j < \alpha. \tag{57}$$

The only pairs allowed to attain the critical slope α while satisfying $Y \log M = X \log A$ should be $(2^m, 3^m)$.

Open closing step

If Target conjecture 8.4 were proved, a noninteger solution would yield the exact asymptotic $\lambda_j \to \alpha$ and simultaneously the strict improvement (57), a contradiction. The desired theorem must be *base-sensitive*: the known integer solutions $(M, A) = (2^m, 3^m)$ attain equality at the same slope and cannot satisfy a universal strict improvement.

The available classical theorems miss this target in distinct ways.

Input	What it supplies here	Why it does not close the argument
Baker linear independence	Excludes rational quadratic forms of rank at most two at independent prime logarithms.	The determinant is bilinear in logarithms and has rank three or four in the remaining branches.
Six Exponentials	At least one U_j^{ϑ} in every triple is transcendental.	It is qualitative and gives no lower bound for the relative error at slope α.
Classical linear forms in logarithms	Effective lower bounds when coefficients are algebraic and the expression is linear in logarithms.	Here the "coefficients" X, Y are themselves logarithms, producing a 2×2 logarithmic determinant.
Qualitative matrix-of-logarithms rank theorems	Nonvanishing and rank restrictions under suitable support hypotheses.	They do not quantify how close a moving matrix can be to rank one as its entries and supports grow.
Subspace-theorem gcd bounds	Subexponential bounds for gcds such as $\gcd(M^n - 1, A^n - 1)$ when M, A are multiplicatively independent.	The elementary private- prime theorem guarantees only $q_j \geq 2^{j+1} + 1$; subexponential gcd control still permits shared prime mass of size $\exp(o(2^j))$ and does not force cross-family disjointness.

9 A sharp fixed-polynomial contact ceiling

The dyadic binomial may look arbitrary. The next theorem shows that it is extremal among all fixed monic polynomial transfers.

Theorem 9.1 (Fixed-polynomial contact law)**.** *Let $P \in \mathbb{Z}[T]$ be monic of degree $d \geq 1$. Suppose its first nonleading term is*

$$P(T) = T^d + cT^{d-r} + O(T^{d-r-1}), \qquad c \neq 0, \qquad 1 \leq r \leq d. \tag{58}$$

For all sufficiently large n, $P(M^n)$ and $P(A^n)$ are positive. Define

$$\Delta_{P,n} = \mathscr{D}(P(M^n), P(A^n)). \tag{59}$$

Then

$$\Delta_{P,n} = cYM^{-rn} - cXA^{-rn} + O(M^{-(r+1)n}), \tag{60}$$

so, in particular,

$$\Delta_{P,n} \sim cYM^{-rn}. \tag{61}$$

Relative to the output height,

$$\lim_{n\to\infty} \frac{-\log|\Delta_{P,n}|}{\log P(A^n)} = \frac{r}{d\vartheta} \le \frac{1}{\vartheta} = \alpha. \tag{62}$$

Equality holds if and only if

$$P(T) = T^d + c. \tag{63}$$

Proof. Factor the leading term:

$$P(T) = T^d \left(1 + cT^{-r} + O(T^{-r-1})\right).$$

Because $2r \ge r+1$, the logarithm has the expansion

$$\log P(T) = d\log T + cT^{-r} + O(T^{-r-1}). \tag{64}$$

Substitute $T = M^n$ and $T = A^n$. The leading terms cancel by $Y\log M = X\log A$, giving (60). Since $A^{-rn} = M^{-r\vartheta n} = o(M^{-rn})$, the main asymptotic follows. Finally,

$$-\log|\Delta_{P,n}| = rn\log M + O(1), \qquad \log P(A^n) = dn\log A + o(n),$$

which yields (62). Since $1 \le r \le d$, the slope is at most $1/\vartheta$. Equality forces $r = d$, meaning no term of degree $d-1, \dots, 1$ occurs; thus $P(T) = T^d + c$. The converse is immediate. ☐

New rigorous result in this continuation

The theorem is a rigorous no-go for a broad family of approximation ideas. No fixed polynomial with several lower terms can beat the critical dyadic slope. To improve the argument one must use changing arithmetic support, an adelic input, or a theorem that classifies the equality case rather than merely optimizing the analytic expansion.

Cyclotomic polynomials make the ceiling completely explicit.

Lemma 9.2 (First lower cyclotomic term)**.** *Let $m \ge 2$, $R = \mathrm{rad}(m)$, and $h = m/R$. Then*

$$\Phi_m(T) = \Phi_R(T^h), \qquad \deg\Phi_m = h\varphi(R), \tag{65}$$

and

$$\Phi_m(T) = T^{h\varphi(R)} - \mu(R)T^{h(\varphi(R)-1)} + O\left(T^{h(\varphi(R)-2)}\right), \tag{66}$$

with the evident interpretation when $\varphi(R) = 1$.

Proof. The identity (65) is the standard reduction of a cyclotomic polynomial to its squarefree kernel. For squarefree R, the sum of the primitive Rth roots of unity is $\mu(R)$; therefore the coefficient below the leading coefficient of Φ_R is $-\mu(R)$. Substitute T^h. □

Corollary 9.3 (Cyclotomic optimization)**.** *For fixed $m \geq 2$,*

$$\mathscr{D}(\Phi_m(M^n), \Phi_m(A^n)) \sim -\mu(R) Y M^{-hn}, \tag{67}$$

where $R = \mathrm{rad}(m)$ and $h = m/R$. Its height-normalized contact slope is

$$\frac{1}{\vartheta\varphi(R)}. \tag{68}$$

The maximum possible value is $1/\vartheta$, and among $m \geq 2$ it occurs exactly when $R = 2$, i.e. when m is a power of two.

Proof. Apply Theorem 9.1 with $d = h\varphi(R)$, $r = h$, and $c = -\mu(R)$. Since R is squarefree and at least 2, $\varphi(R) \geq 1$, with equality only for $R = 2$. □

For $m = p^a$, the factor is a p-term geometric sum and its defect is positive by Theorem 5.3. The dyadic case $p = 2$ is therefore simultaneously:

- a positive entropy increment;
- a factor with pairwise disjoint odd support across levels;
- the unique cyclotomic family attaining the critical slope.

This triple optimality is the main reason to prioritize the dyadic ladder.

10 The rank-three external-ray certificate

This section uses one structural statement from the attached unified report. In its rank-three branch, the exponent vectors at all primes outside $\{2, 3\}$ lie on one rational ray. After taking the primitive integer direction of that ray, there is an integer $C > 1$ with $\gcd(C, 6) = 1$ and nonnegative integers

$$m_2, m_3, a_2, a_3, r, s \tag{69}$$

with $r + s > 0$ such that

$$M = 2^{m_2} 3^{m_3} C^r, \qquad A = 2^{a_2} 3^{a_3} C^s. \tag{70}$$

The case of a single external prime is obtained by taking $C = p$.

Put

$$Z = \log C. \tag{71}$$

Then X, Y, Z are linearly independent over $\mathbb{Q}$. Expanding $Y \log M - X \log A = 0$ gives the ternary quadratic relation

$$Q(X, Y, Z) = -a_2 X^2 + (m_2 - a_3) XY + m_3 Y^2 + (rY - sX) Z = 0. \tag{72}$$

Twice this quadratic form has symmetric matrix

$$H = \begin{pmatrix} -2a_2 & m_2 - a_3 & -s \\ m_2 - a_3 & 2m_3 & r \\ -s & r & 0 \end{pmatrix}. \tag{73}$$

Theorem 10.1 (Nonzero integer rank certificate). *Define*

$$D = a_2 r^2 - (m_2 - a_3) rs - m_3 s^2 \in \mathbb{Z}. \tag{74}$$

Then

$$\det H = 2D \tag{75}$$

and

$$D \neq 0. \tag{76}$$

Furthermore, the exact identity

$$DXY = (rY - sX)\big(a_2 rX + m_3 sY + rsZ\big) \tag{77}$$

holds.

Proof. A direct determinant expansion gives (75). The quadratic form (72) is nonzero because $r + s > 0$. If $D = 0$, then H has rank at most 2. Over $\overline{\mathbb{Q}}$, a quadratic form of rank at most 2 factors into two linear forms. Its vanishing at the $\mathbb{Q}$-independent logarithms X, Y, Z would then give a nontrivial algebraic linear relation among those logarithms, contradicting Baker's homogeneous linear-independence theorem. Hence $D \neq 0$.

For the factorization, a direct expansion gives the polynomial identity

$$\begin{aligned} &(rY - sX)(a_2 rX + m_3 sY + rsZ) - DXY \\ &= rs\, Q(X, Y, Z). \end{aligned} \tag{78}$$

Equation (72) now proves (77). The same formula remains valid when $r = 0$ or $s = 0$. ☐

Corollary 10.2 (Edge branches and sign). *Under the hypotheses above:*

(i) *if $r > 0$ and $s = 0$, then $a_2 > 0$, hence $2 \mid A$;*

(ii) *if $r = 0$ and $s > 0$, then $m_3 > 0$, hence $3 \mid M$;*

(iii) *if $r, s > 0$, then*

$$\operatorname{sgn} D = \operatorname{sgn}(r \log 3 - s \log 2) = \operatorname{sgn}(r\vartheta - s). \tag{79}$$

Proof. If $s = 0$, then $D = a_2 r^2 \neq 0$, so $a_2 > 0$. If $r = 0$, then $D = -m_3 s^2 \neq 0$, so $m_3 > 0$. If $r, s > 0$, the second factor in (77) is positive, as are X, Y. ☐

Because D is a nonzero integer, (77) also yields a concrete approximation lower bound.

Corollary 10.3 (External-ray approximation bound). *If $r, s > 0$, then*

$$|rY - sX| = \frac{|D| XY}{a_2 rX + m_3 sY + rsZ} \geq \frac{XY}{r \log A + s \log M}. \tag{80}$$

Equivalently,

$$|r\vartheta - s| \geq \frac{Y}{r \log A + s \log M} = \frac{\vartheta}{x(r\vartheta + s)}. \tag{81}$$

Proof. The first equality is (77). Since $|D| \geq 1$ and

$$\begin{aligned} r\log A + s\log M &= a_2rX + a_3rY + rsZ + m_2sX + m_3sY + rsZ \\ &\geq a_2rX + m_3sY + rsZ, \end{aligned}$$

we obtain the inequality. Divide by X and use $\log M = xX$, $\log A = xY$, and $Y = \vartheta X$. □

Remark 10.4 (Interpretation)**.** The certificate D is the exact integer obstruction preventing the rank-three quadratic form from degenerating to rank two. Its sign chooses the side from which s/r approximates ϑ, and its nonzero integrality gives a quantitative bound without any unspecified constant. The bound still contains the unknown x and is not strong enough to contradict continued-fraction approximation; it is a branch invariant, not a proof.

New rigorous result in this continuation

The rank-three reduction in the unified report leaves a split ternary conic. The new factorization (77) turns its nondegeneracy into a scalar integer certificate and separates the edge cases $r = 0$ and $s = 0$. This is suitable for a short Lean formalization and for branch-specific searches.

11 Stress tests and exact failure points

A useful progress report should try to break its own strongest-looking ideas. The following tests prevent the cyclotomic construction from being mistaken for a proof.

11.1 Integer solutions are exact controls

For every integer $m \geq 1$, the pair

$$M = 2^m, \qquad A = 3^m \tag{82}$$

satisfies every identity in Sections 4 to 7. In particular:

- all defects have the same signs;
- every prime-power partition is positive;
- the dyadic factors have fresh private primes;
- the moving logarithm matrices have critical slope α;
- Six Exponentials still forces transcendence among triples of the numbers $(2^{m2^j} + 1)^{\vartheta}$.

Thus none of those properties alone distinguishes integral from nonintegral x.

Failure audit 11.1 (Control-family obstruction)**.** Any proposed determinant lower bound that is uniform over all M, A and improves the critical exponent below α is false, because it would also exclude the genuine integer pairs (82). The arithmetic hypothesis must explicitly use simultaneously power-primitive external support or must classify equality cases.

11.2 Fresh primes do not yet couple the two ladders

The dyadic gcd law controls

$$\gcd(U_i, U_j), \qquad \gcd(V_i, V_j)$$

for distinct levels, but not $\gcd(U_i, V_j)$. If an odd prime $q \mid U_i$, then $\operatorname{ord}_q(M) = 2^{i+1}$. If also $q \mid V_j$, then $\operatorname{ord}_q(A) = 2^{j+1}$. The real relation $\log A/\log M = \vartheta$ has no immediate meaning in the finite cyclic group $(\mathbb{Z}/q\mathbb{Z})^{\times}$. Bridging that mismatch is precisely an adelic problem.

Known subspace-theorem estimates imply that for multiplicatively independent fixed integers M, A, the gcd of $M^n - 1$ and $A^n - 1$ is subexponential in n. The elementary private-prime theorem gives only the lower bound $q_j \geq 2^{j+1} + 1$ and does not force a prime of exponential size in 2^j. A subexponential gcd estimate therefore still permits shared primes, or shared products of primes, of size $\exp(o(2^j))$.

11.3 Qualitative transcendence is too coarse

Six Exponentials proves that at least one U_j^{ϑ} in every triple is transcendental, but the desired contradiction concerns the size of

$$\vartheta \log U_j - \log V_j.$$

A transcendental number may be extremely close in relative terms to an integer. The classical theorem provides no exponent, no dependence on prime support, and no equality classification at the slope α.

11.4 p-adic order information is one-sided

At a private prime $q_j \mid U_j$,

$$M^{2^j} \equiv -1 \pmod{q_j}, \qquad \operatorname{ord}_{q_j}(M) = 2^{j+1}.$$

This is exact and strong, but q_j does not divide M, so the congruence does not directly interact with the prime-exponent vector of M. A successful p-adic step would need to transport the real near-proportionality of the two log columns into a relation involving A modulo q_j or a common resultant. No such transport is currently proved.

11.5 The polynomial ceiling blocks a common escape

Replacing $T^{2^j} + 1$ by a more elaborate fixed polynomial cannot improve the normalized contact exponent. Theorem 9.1 proves that lower polynomial terms only reduce the slope, and Corollary 9.3 shows that all non-dyadic cyclotomic families are worse by the factor $\varphi(\operatorname{rad} m)$.

11.6 What remains genuinely plausible

After the audits, three nonredundant directions remain.

1. **Cross-ladder adelic control.** Use the private order conditions at divisors of U_j and V_k together with resultants or support-sensitive matrix-logarithm bounds.

2. **Equality-case classification.** Prove that critical determinant decay with prime support outside $\{2, 3\}$ is impossible, while explicitly allowing $(2^m, 3^m)$.
3. **Rank-three branch elimination.** Combine the integer certificate D with additional local conditions on the external integer C. The exact factorization is a smaller target than the full rank-four problem.

12 Numerical calibration and reproducibility

Numerics are included only to calibrate the exact asymptotics. They are performed on the known integer control $x = 1$, where $M = 2$ and $A = 3$. Let

$$R_j = \frac{\Gamma_j M^{2^j}}{Y}, \qquad \lambda_j = \frac{-\log \Gamma_j}{\log V_j}. \tag{83}$$

The theorems predict $R_j \to 1$ and $\lambda_j \to \alpha$.

j	2^j	Γ_j	R_j	λ_j
1	2	$1.7211790324027 \times 10^{-1}$	0.626673868536	0.764173953182
2	4	$5.8097974438167 \times 10^{-2}$	0.846128885139	0.645746715285
3	8	$4.1774557447219 \times 10^{-3}$	0.973435926104	0.623281520363
4	16	$1.6747263074177 \times 10^{-5}$	0.999031818732	0.625634482844
5	32	$2.5579023220339 \times 10^{-10}$	0.999998537502	0.628254606989
6	64	$5.9555891504460 \times 10^{-20}$	0.999999999997	0.629592159480
7	128	$3.2285313476833 \times 10^{-39}$	1.000000000000	0.630260956526

Table 3: Dyadic determinant calibration for the integer control $x = 1$. The limiting value is $\alpha = 0.630929753571457\ldots$.

The nonmonotonic early behavior of λ_j is harmless: the denominator $\log(3^{2^j} + 1)$ and the constant in $-\log \Gamma_j$ still influence small levels. The normalized amplitude R_j rapidly confirms the explicit bound (44).

The following short script reproduces the table without subtracting large nearly equal quantities; it uses `log1p` on the small tails.

Listing 1: High-precision numerical calibration.

```
import mpmath as mp

mp.mp.dps = 100
x = mp.mpf(1)
M = mp.power(2, x)
A = mp.power(3, x)
X = mp.log(2)
Y = mp.log(3)
alpha = X / Y

for j in range(1, 8):
    r = 2**j
    gamma = Y*mp.log1p(M**(-r)) - X*mp.log1p(A**(-r))
```

```
    logV = r*mp.log(A) + mp.log1p(A**(-r))
    normalized = gamma*M**r/Y
    slope = -mp.log(gamma)/logV
    print(j, r, gamma, normalized, slope)

print("alpha =", alpha)
```

A symbolic check of the rank-three identity can be reduced to

Listing 2: Symbolic verification of the certificate identity.

```
import sympy as sp
X,Y,Z,a2,m2,a3,m3,r,s = sp.symbols(
    "X Y Z a2 m2 a3 m3 r s")
Q = (-a2*X**2 + (m2-a3)*X*Y + m3*Y**2
     + (r*Y-s*X)*Z)
D = a2*r**2 - (m2-a3)*r*s - m3*s**2
R = (r*Y-s*X)*(a2*r*X + m3*s*Y + r*s*Z)
assert sp.factor(R - D*X*Y - r*s*Q) == 0
```

13 Lean formalization blueprint

The new exact layer is deliberately elementary enough to formalize before attempting the open quantitative theorem. A modular plan is given in Table 4.

Proposed module	Principal theorem statements	Main Mathlib dependencies
`ShiftedLog Defect`	Exact u-formulas, signs of $\Delta_n^{\pm}$, derivative of E, dyadic identity, explicit bound, asymptotics.	`Real.log`, derivatives, real powers, monotonicity, finite/infinite geometric limits.
`GeometricQuotient Defect`	$\mathcal{G}_{k,n} = E_n - E_{kn} > 0$ and the finite-sum convexity proof.	Finite sums, `Real.rpow`, strict convexity or the power-sum inequality.
`RenyiCyclotomic Defect`	Exact Rényi identity, prime-power cyclotomic partition, telescoping sum and product.	Cyclotomic identities already in algebraic number-theory libraries; analytic convergence.
`DyadicPrivate Prime`	Exact gcd law, existence of an odd divisor, order 2^{j+1}, congruence $q \equiv 1$, privacy.	`Nat.exists_prime_and_dvd`, modular powers, multiplicative order, elementary parity.
`DyadicLog Independence`	Multiplicative independence and finite rational independence of the logarithms.	Prime valuations, unique factorization, clearing rational denominators.
`MovingLog Matrix`	Determinant sign, distance-to-line formula, critical slope, relative/absolute error split.	2×2 matrices, Euclidean norm, elementary limits.

Proposed module	Principal theorem statements	Main Mathlib dependencies
`PolynomialContact Ceiling`	Log expansion for a fixed monic polynomial and the slope $r/(d\vartheta)$.	Polynomial degree/coefficient API, eventual positivity, Taylor estimates for `Real.log`.
`ExternalRay Certificate`	$\det H = 2D$, polynomial identity $R - DXY = rsQ$, edge cases and sign consequences.	Pure ring normalization (`ring`), integer nonzero arithmetic; Baker remains an explicit imported mathematical hypothesis unless a suitable theorem is already formalized.

Table 4: A proposed Lean decomposition.

13.1 Suggested theorem signatures

The finite algebraic core can be separated from analytic assumptions. Schematic Lean interfaces are:

Listing 3: Schematic Lean interfaces, not compiled code.

```
-- Pure determinant/cyclotomic algebra
 theorem shifted_add_eq_minus_double
   (M A n : Nat) (hM : 1 < M) (hA : 1 < A) :
   deltaPlus M A n + deltaMinus M A n = deltaMinus M A (2*n)

 theorem dyadic_gcd
   (T i j : Nat) (hT : 1 < T) (hij : i != j) (hi : 1 <= i) (hj : 1 <= j) :
   Nat.gcd (T^(2^i) + 1) (T^(2^j) + 1) = if Even T then 1 else 2

 theorem externalRay_ring_identity :
   (r*Y-s*X)*(a2*r*X+m3*s*Y+r*s*Z) - D*X*Y = r*s*Q

-- Analytic layer under the exact base-log relation
 theorem deltaPlus_pos
   (hrel : Real.log 3 * Real.log M = Real.log 2 * Real.log A) :
   0 < deltaPlus M A n
```

The exact gcd and ring identities should be formalized first: they are short, stable, and independent of transcendence infrastructure. The next priority is the sign theorem for Δ_n^+, because it packages the strict power-sum inequality that also proves all prime-power cyclotomic increments positive.

13.2 Formalization will not manufacture the missing theorem

Lean can certify every exact identity and prevent silent sign or index errors. It cannot replace Target conjecture 8.4. A faithful final theorem should expose any new matrix-logarithm or transcendence estimate as an explicit hypothesis until a proof is available. The report's recommended workflow is therefore:

1. formalize the elementary ladder and certificate modules;
2. use them to state the critical-slope theorem with no hidden assumptions;

3. search for the missing adelic input while the exact interface remains fixed;
4. only then connect the new theorem to the final contradiction.

14 Prioritized research program

14.1 Priority A: cross-ladder resultants

For $i, j \geq 1$, study

$$G_{i,j} = \gcd(M^{2^i} + 1, A^{2^j} + 1). \tag{84}$$

An odd prime $q \mid G_{i,j}$ imposes simultaneous exact orders

$$\mathrm{ord}_q(M) = 2^{i+1}, \qquad \mathrm{ord}_q(A) = 2^{j+1}. \tag{85}$$

Three concrete questions are now sharply posed.

1. Can a support-sensitive resultant bound force most private prime mass of U_i to remain outside every V_j in a controlled range?
2. Does the simultaneously power-primitive external support of (M, A) imply a lower bound for the product of private prime parts not shared by the opposite ladder?
3. Can (85) be combined with a p-adic logarithm estimate after passing to suitable powers that are 1 modulo q?

A negative answer at one level is not enough; the determinant target needs an aggregate bound over infinitely many levels.

14.2 Priority B: equality cases in matrix-logarithm estimates

Rather than seek a stronger generic lower bound, seek a theorem with a sharp exceptional set. A model statement is:

> If an integer pair (M, A) has the dyadic fresh-support matrices (42) at critical slope α, then either $(M, A) = (2^m, 3^m)$ or its external prime-exponent tensor satisfies an explicit exceptional algebraic configuration.

The current rank-three certificate suggests that the exceptional configuration may be expressed through the integer D and the external integer C. Even a theorem reducing all critical pairs to finitely many ray types would be substantial progress.

14.3 Priority C: exploit all prime partitions simultaneously

For every prime p, the same E_n has the positive partition

$$E_n = \sum_{j \geq 1} \mathcal{C}_{p^j, n}.$$

The equality of the p- and q-partitions is analytically automatic, but their factor supports differ. A genuinely adelic invariant could compare the prime ideals appearing in these two factorizations before taking logarithms. The dyadic partition is optimal in contact, while an odd-prime partition has more terms per factor and different order constraints; using both may prevent the integer-control symmetry from remaining invisible.

14.4 Priority D: finish the rank-three branch first

In rank three, write M, A as in (70). The following finite data are available:

$$(m_2, m_3, a_2, a_3, r, s, D, C), \qquad D \neq 0, \qquad \operatorname{sgn} D = \operatorname{sgn}(r\vartheta - s). \tag{86}$$

A branch search should combine:

- exact local restrictions on C and on the parity of M, A;
- continued-fraction information on s/r;
- primitive-output conditions from the unified report;
- private-prime orders in $M^{2^j} + 1$ and $A^{2^j} + 1$.

If rank three can be eliminated, the remaining rank-four problem will have a clearer statement: genuinely two-dimensional external support is necessary.

15 Conclusion

The continuation has produced an exact new architecture rather than a claimed final proof. Starting from a hypothetical integer point (M, A) on $A = M^{\log_2 3}$, one obtains:

1. signed logarithmic defects at $M^n \pm 1$ and $A^n \pm 1$;
2. positive telescoping decompositions through every prime-power cyclotomic ladder;
3. an exact Rényi-entropy formula for each increment;
4. a dyadic family whose odd prime supports are pairwise disjoint;
5. fresh-support 2×2 logarithm matrices approaching rank one at the exact critical slope $\alpha = \log_3 2$;
6. a proof that fixed polynomial and cyclotomic transfers cannot improve that slope;
7. a nonzero integer certificate controlling the inherited rank-three external-ray branch.

The strongest conceptual advance is the separation of two phenomena that previously sat inside a single zero determinant:

$$\text{analytic critical contact} \quad + \quad \text{level-by-level fresh arithmetic support.} \tag{87}$$

The analytic contact alone is shared by all integer solutions. The fresh support alone is also shared by all integer solutions. The missing theorem must couple the two while recognizing $(2^m, 3^m)$ as the exact equality family.

Open closing step

The most precise current target is Target conjecture 8.4: prove a strict height-normalized determinant improvement for simultaneously power-primitive pairs with external support, or classify all equality cases. This is narrower than a generic four-exponentials theorem, because it concerns a very structured sequence of moving matrices with pairwise private dyadic supports. It is also properly scoped: no currently cited theorem supplies the required quantitative, support-sensitive conclusion.

A A compact proof ledger

Statement	Status	Dependency
Signs and monotonicity of $\Delta_n^{\pm}, E_n$	NEW	Elementary real analysis
Positive dyadic and prime-power partitions	NEW	Factorization and telescoping
Rényi identity	NEW	Algebraic normalization of a finite sum
Dyadic exact gcd/private order	NEW	Modular arithmetic and parity
Logarithmic independence of dyadic factors	NEW	Unique factorization
Fresh-support near-rank-one theorem	NEW	Previous elementary results
Transcendence in every triple	CLASSICAL	Six Exponentials
Polynomial contact ceiling	NEW	Fixed polynomial log expansion
Cyclotomic optimum	NEW	Squarefree-kernel identity for Φ_m
External-ray reduction	INHERITED	Attached unified report
Nonzero certificate D	NEW + CLASSICAL	Determinant algebra plus Baker
Support-sensitive critical-slope rigidity	OPEN	Missing quantitative theorem

Table 5: Proof and trust ledger.

B Selected derivations in expanded form

B.1 The dyadic infinite product

For finite J,

$$\prod_{j=0}^{J}(1+t^{2^j}) = \frac{1-t^{2^{J+1}}}{1-t}.$$

Apply this first with $t = M^{-n}$ and then with $t = A^{-n}$. Raising the first identity to ϑ, dividing by the second, and letting $J \to \infty$ gives

$$\prod_{j=0}^{\infty} \frac{(1 + M^{-2^j n})^{\vartheta}}{1 + A^{-2^j n}} = \frac{1 - A^{-n}}{(1 - M^{-n})^{\vartheta}}.$$

The large powers cancel inside each ratio because

$$\frac{(M^{2^j n} + 1)^{\vartheta}}{A^{2^j n} + 1} = \frac{(1 + M^{-2^j n})^{\vartheta}}{1 + A^{-2^j n}}.$$

This proves (50) without interchanging a conditionally convergent series: every logarithmic term is positive and the finite products telescope exactly.

B.2 The entropy packet at a prime-power level

Let $h = p^{j-1} n$ and $s = M^{-h}$. Reverse the p terms in $\Phi_{p^j}(M^n)$ and normalize:

$$\Phi_{p^j}(M^n) = M^{(p-1)h}(1 + s + \cdots + s^{p-1}).$$

The corresponding A factor is

$$\Phi_{p^j}(A^n) = A^{(p-1)h}(1 + s^{\vartheta} + \cdots + s^{(p-1)\vartheta}).$$

The leading terms cancel in $\mathscr{D}$, leaving

$$\frac{\mathcal{C}_{p^j,n}}{X} = \vartheta \log(1 + s + \cdots + s^{p-1}) - \log(1 + s^{\vartheta} + \cdots + s^{(p-1)\vartheta}).$$

If

$$\pi_\ell = \frac{s^\ell}{1 + s + \cdots + s^{p-1}},$$

then this is

$$-\log \sum_{\ell=0}^{p-1} \pi_\ell^{\vartheta} = (\vartheta - 1) H_{\vartheta}(\pi).$$

The first correction comes from $\ell = 1$, so $\mathcal{C}_{p^j,n} \sim Ys = Y M^{-p^{j-1} n}$.

B.3 Why dyadic factors are optimally tangent

For a fixed monic P, the first lower term determines the first nonzero perturbation of $\log P(T)$ from $d \log T$. A gap of r degrees creates a tail T^{-r}, while the height grows like $d \log T$. Their ratio is therefore r/d. The largest possible gap is $r = d$, when the only lower term is constant. A dyadic cyclotomic polynomial is exactly such a binomial:

$$\Phi_{2^a}(T) = T^{2^{a-1}} + 1.$$

Every cyclotomic polynomial whose squarefree kernel contains an odd prime has at least $\varphi(R) \geq 2$ primitive-root directions, reducing the normalized gap by that factor.

C Bibliography and current-source ledger

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